

Hengguang Zhou

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RESEARCH INTERESTS

Multimodality · Reasoning · Agents · Reinforcement Learning

My research centers on advancing **Multimodal Large Language Model (MLLM) post-training**, with a focus on **Multimodal** and **Reasoning/Agents**.

- **Multimodal**: Exploring how models comprehend and interpret the physical world across diverse sensory (VisualThinker, MOSSBench).
- **Reasoning/Agents**: Eliciting the reasoning and agentic capabilities of (M)LLMs via RL (VisualThinker).

I built **VisualThinker**, the first open-source framework to replicate the “Aha Moment” of Deepseek-r1 in multimodal, using only a 2B non-SFT model (**100+ citations, 600+ GitHub stars**).

I currently lead **TurningPointAI**, a research team with members and advisors from UCLA, UMD, and PSU, focused on harnessing the power of Multimodal Reasoning.

EDUCATION

University of California, Los Angeles 2025-present
Ph.D. Student in Computer Science (First Year).
Advisor: Prof. Cho-Jui Hsieh
Honors: Amazon Trainium Research Fellowship 2025-2026

University of California, Los Angeles 2023 – 2025
M.S. in Computer Science
Advisor: Prof. Cho-Jui Hsieh

University of Toronto 2016 – 2021
Honours Bachelor of Science with high Distinction in Computer Science
Honors: Dean’s List Scholar for Summer 2017, Winter 2018 and Winter 2019

RESEARCH AND INDUSTRY EXPERIENCE

TurningPointAI, University of California, Los Angeles 2023-present
Lead
Mentor: Dr. Ruochen Wang and Prof. Cho-Jui Hsieh
Project: Multimodal Reasoning (ICLR 2026 submission), Multimodal Oversensitivity (ICLR 2025), Text-to-Image Reward Hacking (CVPR 2026 submission)

DeepSE, Hong Kong University of Science and Technology

2022-2023

Research Assistant

Mentor: Minsoo Khang and Dr. Lucy Park

Project: Ontology-free Visually Rich Document Understanding

Dynamic Graphics Project, University of Toronto

2020

Research Assistant

Mentors: Dr. Hsueh-Ti Liu and Prof. Alec Jacobson

Project: Mesh Convolution Neural Network

HMI Lab, Huawei Technologies Canada

2019-2020

Intern Research Engineer

Mentors: Kevin Fan

Project: AR Mobile Device Projection System for Foldable Interaction Studies (MobileHCI 2020), VR Video Editing

iMoon-Lab, Tsinghua University

2018

Research Intern

Mentors: Zhengyue Huang and Prof. Yue Gao

Project: Camera Constraint-Free Multi-View Convolutional Neural Network (AAAI 2019)

PUBLICATIONS

Zhou, H., Li, X., Wang, R., Cheng, M., Zhou, T., & Hsieh, C. J. (2025). **R1-Zero's "Aha Moment" in Visual Reasoning on a 2B Non-SFT Model.** (*ICLR 2026 Efficient Spatial Reasoning Workshop*). (100+ citations, 600+ GitHub stars)

Chen, Z., Zhang, Y., **Zhou, H.**, Ding, Z., Sun, Y., & Hsieh, C. J. (2025). **Rethinking RL Evaluation: Can Benchmarks Truly Reveal Failures of RL Methods?** (*Submission to ACL 2026*).

Hong, Y., Kao, K.C., **Zhou, H.**, & Hsieh, C. J. (2025). **Understanding Reward Hacking in Text-to-Image Reinforcement Learning.** (*CVPR 2026 Findings*).

Li, X.*, **Zhou, H.***, Wang, R., Zhou, T., Cheng, M. and Hsieh, C.J., (2024). **Mossbench: Is your multimodal language model oversensitive to safe queries?** *ICLR2025*.

Huang, Z., Zhao, Z., **Zhou, H.**, Zhao, X. and Gao, Y., (2019). **Deepccfv: Camera constraint-free multi-view convolutional neural network for 3d object retrieval.** In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 33, No. 01, pp. 8505-8512).

Saniee-Monfared, G., Fan, K., Xu, Q., Mizobuchi, S., **Zhou, L.**, Irani, P.P. and Li, W., (2020). **Tent mode interactions: Exploring collocated multi-user interaction on a foldable device.** In *22nd International Conference on Human-Computer Interaction with Mobile Devices and Services* (pp. 1-12).